

Influence of Gait Retraining on Running Economy: A Review and Potential Applications

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ABSTRACT

Running gait retraining has been reported to reduce lower-extremity loads and the risk for some running-related injuries. This has provided evidence that distance running gait can safely be modified in runners. Optimal running gait at a given speed is the product of anthropometric, neuromuscular, fatigue, and external factors. Because running biomechanics are integrally linked with economy, a major determinant of endurance running performance, it is conceivable that gait retraining could improve performance. The evidence investigating the influence of gait retraining on running economy will be evaluated and potential applications discussed.

INTRODUCTION

Over the past several years, gait retraining for the treatment of running-related injuries (RRI) has become more widely accepted. Specifically, gait retraining seems to be a viable strategy for runners dealing with RRI including patellofemoral pain (PFP) syndrome (94) and anterior compartment

syndrome (27). Gait retraining modes vary from simple (e.g., step rate manipulation) to complex (e.g., reducing tibial acceleration [TA]) with strategies that can be clinic- or field-based (93). Numerous publications have critically reviewed these investigations (3,8) and there seems to be mounting clinical acceptance that neuromuscular patterns can be changed in a short amount of time. Investigations evaluating the long-term adoption of running gait changes are limited; however, some evidence suggest that changes can persist for 1 month (21,66) to 3 months (94). With recognition that a runner can actively change their gait for RRI reasons, the question naturally becomes “Can a runner actively change their running gait for performance improvements?”

Running performance is primarily determined by the combined influence of maximal aerobic capacity (9), lactate threshold (82), and running economy (RE) (17). To some extent, all 3 factors can be improved with appropriate training methods; however, RE can vary substantially between runners of similar aerobic capacities indicating that it is a key component in predicting endurance running performance (24). RE is defined as the energy required to

submaximally run a given distance (77) and is assessed by measuring the steady-state oxygen consumption at a given running velocity. Improving RE would likely lead to improvements in race performance (26,77). Runners can be more or less economical depending on velocity, with both physiological and biomechanical factors contributing to RE. Williams and Cavanagh (91) reported that 54% of RE variation can be explained by biomechanical factors. Specifically, the timing and motion of a runner's stride, ground reaction force application, and optimization of elastic energy return can influence RE. After a gait retraining program, RE may be influenced by changes to one or more of these biomechanical factors. Understanding these factors and how they can be modified is central to the question of whether a runner can improve performance through gait improvements.

KEY WORDS:

running form; technique; performance; foot strike; stride rate; form manipulation; form retraining; vertical oscillation; ground contact time

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DETERMINANTS OF RUNNING GAIT

A runner's gait is biomechanically measured in terms of its timing (spatiotemporal), motion (kinematics), and force (kinetics) (Table 1). Ultimately, the adopted gait pattern is the product of the runner's (a) anthropometry, (b) neuromuscular control, (c) fatigue state, (d) external factors, and (e) speed (Figure 1). Anthropometrically, the lower-extremity segments have intrinsic properties (e.g., length, mass, mass distribution, and musculotendon unit stiffness), which define the passive dynamic characteristics and influence the preferred running pattern (11). In addition, soft tissue flexibility and strength will influence joint motion throughout the running cycle. As runners fatigue, their gait adjusts with subsequent changes in spatiotemporal parameters, kinematics, and kinetics (28,51,59). As a result, RE may change throughout a fatiguing run (35,48). External to the runner, factors such as environmental effects (e.g., wind) and foot-ground interaction (through changes in footwear or ground surface) can influence a runner's gait and RE. Finally, as a runner progresses through a range of speeds, their running gait will adjust (68,71) and their RE can change (71,73). In a recent study of 20 male recreational runners tested across 3 speeds (65, 75, and 85%

velocity of maximal aerobic capacity), 25% were most economical at the slowest speed, 20% at the middle speed, and 55% at the fastest speed tested (71). This indicates that runners will be more or less economical depending on what pace they are running (Table 1).

Several investigations have attempted to determine biomechanical metrics that are common in runners with good RE (7,32,43,49,71,83,91) (Table 2). Although it is beyond the scope of this commentary to review all of these potentially modifiable biomechanical factors, the reader is pointed to Moore (60) for a thorough review. Although biomechanical variables explained some (31–54%) of the RE variance in these investigations, a singular and universal most efficient running gait has not emerged and strong biomechanical predictors are elusive. This is likely due to variations in anthropometry, strength, flexibility, and other factors (e.g., running experience) between individuals. In addition, multiple gait patterns can produce a similar RE, indicating that there is not a universal most economical gait at submaximal speeds (53). In a study of 58 runners, Lussiana et al. (53) found similar RE values across 2 groups, with some runners using a greater reliance on elastic energy return (shorter ground contact time [GCT]), and other runners

limiting the upward translation of the center of mass (COM) (longer GCT). This investigation highlights the individuality of running gait and its relationship with RE.

RUNNING GAIT RETRAINING

Running gait retraining seeks to create a new neuromuscular “solution” for an individual's anthropometry, fatigue state, external factors (e.g., footwear), and running speed (Figure 1). Although most of the running gait retraining interventions target RRI (8), several have evaluated its effect on RE in healthy runners through pre-post intervention RE measurements (Tables 3 and 4). In general, these studies had small sample sizes ($n = 6\text{--}28$) and varied both the implementation (e.g., verbal and visual) and duration (1 session to 12 weeks) of the intervention.

INFLUENCE ON RUNNING ECONOMY

Evidence to support the efficacy of running gait retraining to improve RE is lacking. Of the running gait retraining investigations that have measured RE changes, only one study reported a potential benefit (63). Although Morgan et al. (63) demonstrated that step length could be decreased over 3 weeks to improve RE, the study was

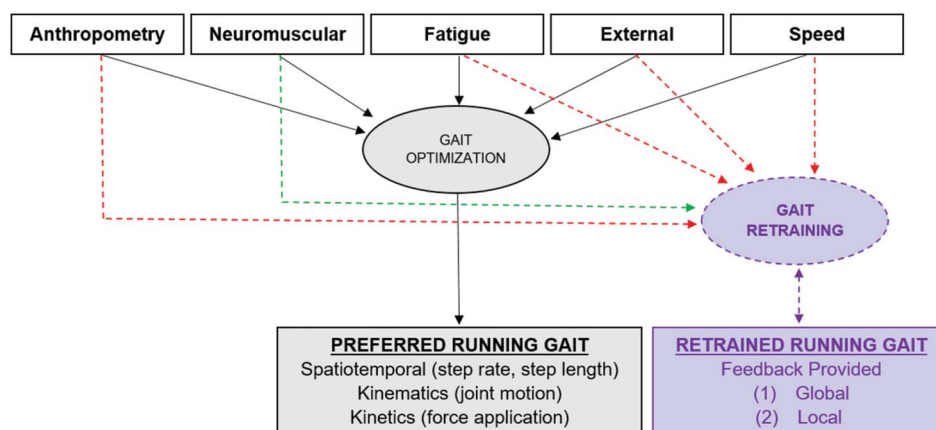


Figure 1. Running gait is the product of several components, including anthropometry, neuromuscular control, fatigue, speed and external factors (e.g., footwear and running surface). Gait retraining specifically seeks to change running form from the preferred/habitual running form by altering the neuromuscular component. Red arrows indicate factors that are not manipulated during a gait retraining program.

Table 1
Definition and common metrics of key running gait biomechanical terms

Term	Definition
Spatiotemporal	The timing and spatial characteristics of the gait cycle. The gait cycle is commonly partitioned into stance (foot on ground) and swing (foot off ground) phases. The product of stride rate and stride length (see below) is equal to running velocity.
Stride rate or step rate ^a	The number of steps or strides (equivalent to 2 steps, defined as successive foot-ground contacts of the same foot) a runner takes in a given amount of time, typically 1 min (e.g., 180 steps per minute or 90 strides per minute). Step rate is sometimes referred to as cadence.
Step length or stride length ^a	The distance covered during a step or stride. Step length is the distance from foot strike of one foot to foot strike of the opposite foot. Stride length is the distance between 2 consecutive foot strikes of the same foot.
Ground contact time ^a	The amount of time the foot is on the ground during a single foot strike (initial foot contact to toe-off). This may also be referred to as “stance time.” Typically assessed with an embedded force plate but can be estimated from a foot accelerometer.
Kinetics	Internal (e.g., muscle forces) or external forces and moments that occur during the gait cycle. The most commonly reported kinetic metric is the reaction forces applied to the foot by the ground, termed ground reaction force (GRF).
Ground reaction force	These forces are typically measured with an embedded force plate in a laboratory setting. Most commonly, the vertical GRF and anterior-posterior GRF are reported. The anterior-posterior GRF is also referred to as the braking and propulsive forces.
Kinematics	The motion of the segments and body during the gait cycle. Typically, joint angles or segment (e.g., thigh) angles are reported.
Vertical oscillation ^a	The change in height of a runner’s center of mass during the gait cycle. Most commonly assessed through motion capture systems or estimated from an accelerometer.
Tibial acceleration	The change in velocity of the tibia bone. This is typically estimated using a skin-mounted accelerometer or motion capture markers and quantifies the shock experienced by the tibia during stance phase.

^aBiomechanical metrics available for both real-time display and post hoc analysis using current consumer wearable technologies.

limited to 9 runners who initially demonstrated uneconomically long step lengths (overstriding). No other interventions were effective at improving RE and no studies have reported a long term follow-up. The longest intervention was 12 weeks and it resulted in a significantly impaired RE in 8 triathletes trained with the Pose method (23). Although many reports highlight the efficacy of running gait retraining in the treatment and prevention of RRI, relatively few have also measured RE pre-post intervention (16,74,85). A gait retraining intervention that limits injury risk without a deleterious effect on RE would be beneficial to performance by mitigating training interruptions. In a group of 22 male runners, Clansey et al. (16) retrained 12 recreational runners to minimize TA and

reported a reduction of impact load without a significant negative effect on RE. Similarly, Roper et al. (74) reported that retraining recreational runners with PFP to use a forefoot strike did not significantly reduce RE. By contrast, Townsend et al. (85) reported that RE was significantly impaired when 12 healthy runners performed a single gait retraining session to lower peak TA. From these reports, RE may be unchanged or compromised when initially altering running gait. When RE is compromised initially, continued practice could return RE to pretraining levels, but long term follow-up reports are needed to evaluate this.

It could be argued that a runner self-optimizes gait and becomes more economical through training. Moore et al. (61) assessed RE in 10 female beginner

runners pre-post a 10-week training program. Although maximal aerobic capacity did not change, RE and treadmill time to exhaustion both improved, which demonstrated that beginner runners improved RE simply through training. Of the variables collected, Moore et al. (61) found that 3 variables explained 94% of the variance in RE: less knee extension at takeoff, delayed peak dorsiflexion in stance, and reduced eversion velocity at foot contact. Nelson and Gregor (65) studied 10 competitive collegiate distance runners over a two-year period and found that they all decreased stride length, increased stride rate, and decreased GCT in the absence of any gait retraining intervention. Because all these parameters are easy to monitor with advances in wearable technology, coaches should consider tracking these metrics. Interestingly, similar findings

Table 2
Summary of potential biomechanical predictors to improved running economy

Investigation	Population	Biomechanical predictors of running economy
Williams and Cavanagh (91)	31 elite male distance runners	More extended leg at foot strike, less extended leg at toe off, reduced peak vertical GRF, forward trunk lean, and more predominate RFS
Kyröläinen et al. (49)	17 competitive collegiate middle distance runners (male & female)	No kinematic predictors; reduced braking GRF and mediolateral GRF
Heise and Martin (42)	16 male recreational runners	Reduced total vertical impulse and reduced net vertical impulse
Heise et al. (43)	16 male recreational runners	Greater negative work at hip, greater positive work at knee, and less negative work at ankle
Tartaruga et al. (83)	16 healthy male 10 K distance runners	Reduced SR, increased SL, and more extended knee at ground contact
Barnes et al. (7)	63 competitive collegiate runners (male & female)	Increased leg stiffness and increased Achilles tendon moment arm
Folland et al. (32)	97 female long-distance runners (varying ability)	Decreased vertical oscillation, more extended knee at ground contact, and decreased horizontal pelvic velocity
Pizzuto et al. (71)	20 male recreational runners	Decreased knee flexion-extension ROM, decreased knee and hip Ab/adduction ROM during stance

GCT = ground contact time; GRF = ground reaction force; RFS = rearfoot strike; ROM = range of motion; SL = stride length; SR = stride rate.

were not reported in two 6-week (34,50) and one 7-week retraining study (6). Lake and Cavanagh (50) reported no RE improvements or biomechanical changes after a 6-week run training program in 15 untrained college males. Study inclusion required that participants complete a 30-minute run, which resulted in a substantially more trained subject pool than in Moore et al. (61). It is possible that no RE improvements or gait changes occurred because the intervention was not long enough. However, Franch et al. (34) reported RE improvements of 3.1, 3, and 0.9%, respectively, in recreationally trained runners who engaged in distance, long-interval, or short-interval training over a 6-week period. No changes were found in spatiotemporal mechanics.

INTERVENTIONS

Gait retraining can either provide global and/or local feedback. With global feedback, the runner is provided an overarching goal, but is not instructed on how to alter gait to achieve the stated goal. For instance, a global

strategy may provide biofeedback on TA and instruct runners to reduce it (16). The runner is not given a specific solution and can adjust gait in any manner to accomplish the goal. Conversely, a local feedback strategy targets a specific gait feature and explicitly attempts to adjust it. For instance, runners with PFP have been provided visual feedback on knee position with instructions to limit its valgus movement during the stance phase (94).

Although generally accepted flaws such as extraneous upper-extremity motion are thought to lead to poor RE, evidence does not definitively suggest that targeting one or more purported flaws will lead to improved RE. Based strictly on the available evidence, it seems that running gait retraining to improve RE is difficult. Although it seems plausible that a runner can safely adopt a new gait, the best outcome would seem to be that RE was not degraded. Because there does not exist a universal “optimal” running gait, perhaps the lack of positive findings could be attributed to the need for individualized retraining

programs as opposed to a “one size fits all” approach. Although we are not stating that gait retraining cannot assist an individual runner to become more economical, clear knowledge of what biomechanical criteria to correct across many or all runners is lacking. Despite this uncertainty, we will discuss clinically feasible interventions for individuals using both local and global manipulation. With the advent of wearable technology to measure and track cadence, GCT, vertical oscillation (VOSC), and high-speed video becoming standard within most smartphones, it is important for coaches and clinicians to have an improved understanding of these data.

COMBINED APPROACH

Several investigations have used a combined approach (global and local intervention) to retrain gait (20,23,31,57,70,90). Both Petray and Krahenbuhl (70) and Messier and Cirillo (57) reported no RE improvements when runners received individualized gait feedback. Petray and Krahenbuhl (70) provided

Table 3
Long-term (chronic) running gait retraining interventions where running economy was measured pre- and post-intervention

Investigation	Experimental group	Intervention	Feedback	Length, wk	RE effect
Petray and Krahenbuhl (70)	20 10-yo runners	↓ VOSC, information on posture, relaxation and SR	Verbal	11	No change
Messier and Cirillo (57)	11 novice, female runners	↓ VOSC, ↓ excessive arm and trunk motion, SL, knee position in stance and swing	Verbal, video	5	No change
Bailey and Messier (6)	13 novice, male runners	Freely chosen SL versus fixed SL	Audio (metronome)	7	No change
Morgan et al. (63)	9 runners	Freely chosen SL versus mathematically derived optimal SL	Audio, visual	3	3.4% improvement in mathematically derived optimal SL
Dallam et al. (23)	8 triathletes	Pose method	Verbal, drills	12	Worse
Fletcher et al. (31)	8 male recreational runners	Pose method	Verbal, video, written	1	No change
Clansey et al. (16)	12 male recreational rearfoot strike runners	Reduce PTA	Visual	3	No change
Craighead et al. (20)	9 recreational runners	Midstance-to-midstance method	Verbal, video (individual and group setting)	8	No change
Hafer et al. (38)	6 recreational runners with low SR	Increase SR 10%	Audio (metronome, songs) (self-administered)	6	No change
Warne et al. (89)	12 male moderately trained runners	MFS, ↑ SR 10%, upright trunk, and softer landing	One 30-min instructional session	8	No change
Roper et al. (74)	8 RFS recreational runners	Switch to FFS	Verbal, visual (mirror)	6	No change

FFS = forefoot strike; MFS = midfoot strike; NRFS = nonrearfoot strike; PTA = peak tibial acceleration; RE = running economy; RFS = rearfoot strike; SL = stride length; SR = stride rate; VOSC = vertical oscillation.

verbal instructions over an 11-week intervention to 10 year olds. Participants were instructed to reduce VOSC and were given general information regarding posture, relaxation, and stride rate. After the intervention, they reported no changes in RE, VOSC, stride rate, and stride length. Messier and Cirillo (57) carefully selected 11 female novice runners who demonstrated at least one gait “flaw.” Retraining consisted of comparing a video of each runner to an experienced runner.

Flaws were explained and pointed out to each runner. The runners then performed 3 treadmill runs per week for 5 weeks and received both verbal and visual cueing. A control group performed the same treadmill runs, but was not provided feedback. Similar to the findings of Lake and Cavanagh (50), the RE and biomechanics of the control group were unchanged after the 5 weeks of treadmill runs. However, the retraining group exhibited significantly longer stride lengths, greater

knee flexion during the stance and swing phases, greater dorsiflexion during the stance phase, and increased VOSC (8.81 ± 0.44 versus 11.07 ± 0.43 cm). All these biomechanical changes were intended, with the exception of VOSC. This highlights the interdependence of biomechanical gait parameters as a change in one may cause an unintended change in another. In addition, 2 members of the experimental group were able to adjust to a rearfoot strike (RFS) as

Table 4
Short-term (acute) running gait retraining interventions where running economy was measured pre- and post-intervention

Investigation	Experimental group	Intervention	Feedback	Length	RE effect
Cavanagh and Williams (14)	10 male runners	SL	Audio (metronome)	1 session	Preferred SL had best RE
Ardigò et al. (4)	8 recreational RFS	RFS versus FFS	Verbal instruction	1 session	No change
Gruber et al. (37)	19 RFS and 18 NRFS	Adjusted foot strike	Verbal instruction	1 session	Worse when switching from RFS to FFS at slow and medium speeds. No change when switching from NRFS to RFS
Connick and Lee (18)	11 male “trained” runners	SR	Audio (metronome)	1 session	Preferred SR had best RE
Townsend et al. (85)	12 male recreational runners	Reduce PTA	Verbal, visual	1 session	Worse
Warne et al. (89)	12 recreational runners	Increase SR 10%	Audio (metronome)	1 session	No change
Copriva (19)	12 recreational runners	VOSC	Video	1 session	Best RE at preferred VOSC
Moore et al. (62)	28 RFS recreational runners	Flatter foot contact	Verbal cues (3 types)	1 session	No change in RE for 2 conditions, worse RE for one condition

FFS = forefoot strike; NRFS = nonrearfoot strike; PTA = peak tibial acceleration; RE = running economy; RFS = rearfoot strike; SL = stride length, SR = stride rate; VOSC = vertical oscillation.

instructed. Despite all these biomechanical changes, there were no significant changes to RE or ratings of perceived effort (RPE).

More recently, 3 gait retraining interventions sought to teach runners the Pose method (23,31) and the midstance-to-midstance running program (20). Both Pose method interventions were effective at adjusting gait; however, the adjustments either resulted in no RE change or a significantly worse RE. Dallam et al. (23) exposed participants to the Pose method with 1 hour of instruction per week, whereas Fletcher et al. (31) used drills over 7 days. Both interventions reduced stride length, increased stride rate, and reduced VOSC. Craighead et al. (20) exposed 9 recreational runners to five 1-hour instructional sessions on the midstance-to-midstance running program, which primarily aims to promote midfoot ground contact. The 8-week intervention was

effective at reducing stride length, increasing stride rate, and reducing maximum knee flexion in stance, but did not change foot contact position or VOSC. More importantly, the intervention did not significantly change RE or RPE.

A combined approach was used by Warne et al. (90) in an 8-week intervention in 12 male moderately trained runners. Runners were instructed over a 30-minute session to increase step rate by 10%, aim for a nonrearfoot ground contact, keep the trunk upright, and land softer. Runners were also instructed to run in minimalist style footwear in addition to runs with conventional midsole footwear. Runners significantly increased step rate and shifted away from an RFS in minimalist style footwear. However, RE was unchanged after the intervention. There could be several reasons that these global retraining interventions were not successful at improving

RE, including differing individual responses cancelling out when averaging pre-post differences. This possibility was postulated by Petray and Krahenbuhl (70), who indicated that participants with the greatest pretreatment VOSC had the greatest reduction, whereas participants with low VOSC pretreatment saw little change to their VOSC. Although the current evidence does not support a global retraining strategy for improving RE, it does suggest that runners can learn a new gait pattern that overrides their self-selected one.

SPATIOTEMPORAL MECHANICS

The timing of the gait cycle is perhaps the easiest biomechanical parameter to assess and thus its relationship to RE has been well documented. Reports are equivocal regarding the influence of spatiotemporal parameters on RE (71,83). Taratuga et al. (83) studied 16 male distance runners and determined

that stride rate and stride length had the greatest relationship to RE, accounting for 28% and 23% of its variance, respectively. It is possible that spatiotemporal factors had the greatest relationship to RE because adjusting stride rate may change the leg's position at foot strike, which could influence both elastic energy return and ground reaction force application. Conversely, Pizzuto et al. (71) reported that stride rate and stride length did not have a strong relationship to RE in their study of 20 recreational runners.

Despite equivocal evidence on the importance of stride rate on RE, retraining a runner's stride rate remains an enticing intervention due to the ease of implementation. Stride rate manipulation can be achieved by using a metronome (e.g., 89), music (selecting songs with the desired beats per minute) (38), or wearable technology (93). Willy et al. (93) demonstrated that runners increased stride rate by monitoring output from wearable technology in 8 over-ground runs. Increasing stride rate by 10% has been reported to effectively reduce joint loads (41,52,92), which would theoretically reduce a runner's inherent RRI risk (78). However, none of these studies assessed RE. Evidence suggest that both experienced and inexperienced runners may self-optimize to the most economical pairing of stride length and stride rate (14,47). In addition, runners demonstrate the ability to maintain the self-optimization of spatiotemporal mechanics even when fatigued (87). To date, a one size fits all "optimal" stride rate/length for improved RE has not been established. When runners adjust their stride rate slightly (<8%) from their preferred, there are nonsignificant changes to RE and timing of muscle contractions (18). Several studies support the notion that small changes (<8%) from the preferred stride rate have minimal deleterious effect on RE (14,18); however, it is unknown what percentage change would be meaningful and influence race performance. In addition, a weak association between stride length and anthropometry (e.g., leg length) complicates the determination

of the optimal stride length (13,84). In the absence of a stronger association, prescription of stride length or stride rate based strictly on anthropometry is not recommended.

Although it seems that runners self-optimize their step length based on given constraints (e.g., anthropometry) and experience (65), de Ruiter et al. (75) reported that experienced runners adopted a preferred stride rate 3% lower than their most economical stride rate, whereas novice runners adopted a preferred stride rate 8% lower. This indicates that these runners slightly overstride (i.e., they adopt a freely chosen stride length longer than their most economical one) and if they increased stride rate, RE might improve. Morgan et al. (63) measured oxygen consumption across a range of stride lengths and determined that 9 of 45 recreational runners (20%) were overstriding and would be more economical with an increased stride rate, thereby reducing stride length. Runners received 15 minutes of visual-audio feedback over 15 sessions in a 3-week period to retrain their gait timing in accordance with an increased stride rate. After retraining, runners significantly reduced stride length and RE significantly improved by 3.4%. Over a 6-week unsupervised intervention, Hafer et al. (38) instructed 5 novice runners (with an initial stride rate between 75 and 80 strides per minute) to increase their stride rate by 10%. Runners increased their stride rate; however, the program had a nonsignificant impact on RE. Potential subjects were only screened for a low stride rate and not for an uneconomical stride rate, which could explain why the intervention did not improve RE. Alternatively, the absence of RE improvement could be explained by the lack of supervision, unlike Morgan et al. (63) who constrained runners to complete all runs at the same velocity.

Based on the current evidence, there is not a "gold standard" stride rate that all runners should target to optimize RE. Although it seems that runners with uneconomical overstriding patterns show the potential to retrain gait to

improve RE, identifying which runners to retrain may be challenging in practice. Runners show the capacity to increase stride rate in both treadmill and in-field retraining, which increases the clinical application. Because it does not seem that small increases in stride rate negatively impact RE, this intervention warrants consideration from a coach as it may reduce a runner's risk of RRI (78).

FOOT STRIKE PATTERN

Foot strike pattern is commonly thought to influence RE, which has led some to suggest that it should be a focus of gait retraining. Although Williams and Cavanagh (91) hypothesized that an RFS might be more economical by allowing the footwear midsole to dissipate forces in early stance, promoting a non-RFS (NRFS) has been used in several investigations (23,31,62,90). Proponents of a NRFS point to the high incidence of elite endurance runners using an NRFS (72); however, this has been questioned by surveys of foot strike patterns during races (39,40). The foot strike pattern adopted by runners may be speed dependent, with some runners using a more anterior foot strike at faster speeds (10). Regardless, most of the elite marathon runners at the 2017 IAAF World Championships used an RFS and there was no difference in the incidence of NRFS runners in the top and bottom 50% of men (39). Other arguments for the use of an NRFS pattern are based in the notion that an NRFS pattern may store and return more elastic energy from the Achilles tendon and arch of the foot (69). One study suggested that using an arch-restricting insole to reduce arch compression (lowering and lengthening of the arch due to loading of the foot) during the stance phase of running increased oxygen costs, which supports the link between arch deformation and RE (80). However, arch deformation and elastic energy storage may not differ between NRFS and RFS patterns (55,88), suggesting there may be little difference in RE between the 2 foot strike patterns.

Several investigations have successfully manipulated foot strike in either an acute (1 session) (62) or chronic intervention (12 weeks) (23). Runners demonstrate the capacity to switch foot strike on instruction; however, there is no evidence that switching from RFS to NRFS or vice versa will improve RE. Direct measurements of RE with runners switching between RFS and NRFS suggest that an RFS may be more economical in habitual RFS runners and that the 2 foot strike patterns are equally economical in habitual NRFS runners (37). This is echoed by Melcher et al. (56), who reported that runners (all habitual RFS) were 5.6% more economical when using an RFS at both the beginning and end of a long treadmill run. Additional studies report that RE does not change between foot strike patterns (4,22,69). Based on the available gait retraining evidence, a coach should not expect an improvement in RE if they instruct a runner to switch habitual foot strike pattern.

VERTICAL OSCILLATION

The body's COM goes through a vertical translation throughout the gait cycle. It peaks during the aerial phase and reaches its lowest point during maximum leg compression in stance phase. This overall vertical displacement is termed VOSC. When VOSC is increased during the running cycle, RE is compromised (19,86). In fact, VOSC has been reported to explain 27.7% of RE variation (32) with smaller oscillation being characteristic of better performing runners (91). VOSC has also been reported to increase 17.3% during an exhaustive treadmill run (33). It seems that a reduced VOSC may be beneficial for RE; however, a normative set of VOSC across a wide population of runners and speeds is currently lacking for informed decision-making.

Evidence supports that runners can retrain VOSC through either verbal (2,86), visual (19), or visual plus auditory cueing (29). This raises the prospect of retraining runners to limit VOSC and improve RE. Copriviza (19) provided visual VOSC feedback to 12 recreational runners under 5 conditions. Runners

were most economical at their preferred VOSC and significantly less economical when VOSC was increased. When runners reduced VOSC to 50% and 75% of the preferred VOSC, RE was not significantly different from the preferred condition. Not surprisingly, when VOSC was decreased, there was a concomitant increase in stride rate. Adams et al. (2) recently showed in 20 recreational runners that VOSC could be reduced by either having the runner increase stride rate or by instructing them to limit VOSC. In both interventions, average VOSC was reduced by 17.1 and 21.7%, respectively. Although both retraining methods significantly increased stride rate and reduced VOSC, they had different effects on GCT. When runners focus on increasing cadence, they reduce VOSC and GCT. When runners focus on reducing VOSC, they increase cadence and increase GCT. Because it has been previously reported that increased leg stiffness has been shown to increase stride rate while limiting both GCT and VOSC (30), it is probable that the interventions had different effects on leg stiffness.

These results raise the possibility of gait retraining to improve RE in a runner with a high VOSC. However, identifying a suitable runner is challenging because no target value for VOSC is currently known. Wearable technology may provide valid and reliable measurements of VOSC (1) with a minimum detectable change of 0.45 cm. This raises the prospect of field-based retraining, but additional information about how VOSC changes across a range of speeds and terrains is needed.

GROUND CONTACT TIME

GCT is defined as the amount of time that the foot spends on the ground. It has been determined that GCT has a significant inverse relationship with leg stiffness (64). A stiffer leg may store more elastic energy within the musculotendon units and ligaments before releasing this energy during the propulsive portion of stance. The scientific evidence has been conflicted as to whether a shorter or longer GCT is associated with an improved RE.

Although some studies have reported an improved RE with decreased GCT (67,76), the majority of studies have either reported no significant relationship (42,49,81,91) or an improvement in RE with increased GCT (15,25). Lussiana et al. (54) recently reported that both shorter GCT (greater reliance on elastic energy, increased VOSC) and longer GCT (limiting VOSC) were effective strategies for economical gait. No reported gait retraining intervention has specifically targeted GCT. However, Adams et al. (2) recently demonstrated that interventions targeting either stride rate or VOSC will significantly change GCT. Based on the available evidence, a coach should not explicitly instruct a runner to decrease or increase GCT with the goal of improving RE.

ARM SWING

Arm swing during running is critical to counterbalance the angular momentum produced by the legs (44) and contributes 5–10% of the body's lift off the ground (46). The arms work in an opposite direction to leg motion and their movement helps to limit rotational contributions from the head and trunk (5). When the arms are intentionally restricted during running by placing them on the head or behind the body, RE is significantly worse (5,86). Although this is intuitive, it highlights that arm swing is important for economical running. Computational modeling has demonstrated that suppressed arm swing decreased vertical ground reaction forces by 10–13% of body weight but increased lateral ground reaction forces by 4–6% of body weight (58). This further confirms the “lifting” action of the arms and their role in mitigating the body's side-to-side motion. Unfortunately, investigations of the most economical arm swing pattern are limited. Hinrichs et al. (46) found no advantage to restricting arm swing to the sagittal plane and reported some frontal plane arm motion in the 10 recreational runners they tested. They concluded that this motion across the body helped to minimize the side-to-side COM movement. Hinrichs (45) found that 2

of 10 subjects demonstrated asymmetrical arm motion that potentially compensated for asymmetries in the rest of the body. Clinically, visual observation of asymmetrical arm action may be a clue for the coach that a runner may have other imbalances that the arms may be compensating for. This is an important consideration for a runner returning from injury or during fatigued running. During a fatiguing run, changes to arm swing would most likely be due to a compensatory strategy as opposed to local arm muscle fatigue (79). It is clear that arm movement is dependent on the motion of other body segments and that a retraining strategy to adjust arm swing would have both kinematic and kinetic consequences. However, the evidence currently does not support adjusting arm swing to improve RE. Because hill running requires increased propulsive vertical ground reaction forces (36) and arm motion has been computed to increase these forces (58), the “lifting” motion of the arms may be beneficial during hill running and could be something that coaches cue runners on.

INDIVIDUALIZED APPROACH

Although it is quite clear that there is no basis for a coach implementing a global gait retraining for an entire team, this does not imply that biomechanical interventions cannot improve RE for individual runners. Morgan et al. (63) provided evidence that this is indeed possible. Of all the reviewed gait interventions, this is the only study that used a combined biomechanical and physiological approach to screen runners. They determined the runner's optimal stride length from an initial metabolic screening and then retrained the runner to adopt a more economical stride pattern. Therefore, it is not surprising that all but one (23) of the other studies reviewed here reported no change to RE after gait retraining. It is quite possible, and likely, that some of the participants in the aforementioned studies did actually experience an improvement of RE that would aid their performance. This is supported by the individual results reported by Gruber et al. (37). Despite the study's statistical

conclusion that an NRFS is not more economical than an RFS, 5–31% (depending on running speed) of habitual RFS runners were more economical when switching to an NRFS. Similarly, 22–26% of habitual NRFS runners were more economical when using an RFS. This implies that adjusting foot strike was effective at improving RE for some runners at certain speeds and the more economical foot strike differs between individuals. Cavanagh et al. (12) provided an extensive biomechanical profile of 2 elite distance runners from the 1984 Los Angeles Summer Olympics and recommended that any gait modification be supported by multiple metabolic measurements. The key ingredient for the success of a gait retraining is determining gait features that a runner can safely adjust that would also result in improved RE. Currently, this would only be possible from a combined biomechanical and physiological approach.

SUMMARY

Based on the currently available literature, there is minimal evidence to support the implementation of a universal running gait retraining strategy for the explicit reason of improving RE and potentially performance. Although a runner's biomechanics have been shown to influence RE, little evidence suggests that biomechanical changes can improve RE. It seems that runners self-optimize gait within a given set of constraints (anthropometric, strength, fatigue, speed, etc.) to produce an economical movement pattern. For elite runners, the prospect of adjusting gait is possible, but physiological measurements should guide decision-making in the determination of which biomechanical parameters to adjust. Both novice and experienced runners have been shown to optimize running gait on the adoption of consistent training and no explicit instruction. Runners have shown the capacity to adjust gait parameters using a variety of retraining strategies and with varying intervention lengths. However, the literature is lacking investigations of the long-term adoption and potential injury risks of these gait patterns. From a clinical

perspective, adjusting a runner's gait to minimize the reoccurrence of a running-related injury seems warranted without a deleterious influence of RE. If such an adjustment could help avoid a running-related injury, a runner could engage with more consistent training which, in turn, could precipitate an improvement in RE. Because wearable technologies are now able to reliably track stride rate, VOSC, and GCTs, coaches are encouraged to monitor these metrics within runners. However, future reports and case studies will be required to establish criterion values to improve RE on an individualized basis.

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